

PROBLEM SET 4

COM SCI 188: Intro to Robotics

Spring 2025

Question:	1	2	3	4	5	Total
Points:	20	30	20	20	10	100

1. Multiple Choices.

- (a) (5 points) In a hospital, a service robot named Moxi delivers supplies and picks up lab samples. What is a key HRI challenge in this setting?
- A) Optimizing battery life
 - B) Interpreting spoken medical jargon
 - C) Navigating without disturbing human workflows
 - D) Operating medical equipment
- (b) (5 points) A robot named Paro looks like a baby seal and is used in therapy for elderly patients. What kind of interaction is it designed to foster?
- A) Technical troubleshooting
 - B) Emotional and social engagement
 - C) Physical rehabilitation
 - D) Financial management
- (c) (5 points) Which robotic system is most likely to involve shared autonomy?
- A) An industrial welding arm
 - B) A robotic vacuum cleaner
 - C) A robotic exoskeleton for paraplegics
 - D) A Mars rover operated remotely
- (d) (5 points) Imagine a robot helps elderly people at home by reminding them to take medication and connecting them with family via video calls. Some users may have mild cognitive decline. Which of the following is the most relevant HRI challenge in this scenario?
- A) Avoiding stairs and obstacles while navigating
 - B) Interpreting complex voice commands in multiple languages
 - C) Modeling communication with users who may have memory issues
 - D) Performing high-speed data processing for autonomous decision-making

2. Short Answers.

- (a) (10 points) A robot is deployed in a museum to interact with visitors, provide information, and guide tours. List two HRI design considerations critical for this environment.

- (b) (10 points) Name one challenge robots face when assisting with activities of daily living (ADL e.g., bathing, dressing) in a home care environment.

- (c) (10 points) What is one major technical limitation that can affect human-robot interaction in teleoperation over long distances?

3. (20 points) **Robot chef.** You're tasked with designing a robot to assist human chefs in a busy industrial kitchen. The robot must collaborate with staff, share workspaces, and adapt to unpredictable human behavior and timing.

What HRI features would you prioritize to support smooth collaboration and turn-taking? Who are the key stakeholders you would consult during the design process, and why?

4. (20 points) **Robot lamp.** Apple has developed a lamp-like robot ELEGNT that combines functional movement with expressive behaviors.

Imagine this expressive robot is integrated into a home or workplace environment. What novel use cases can you envision that leverage its ability to convey intention, attention, or emotion through movement? How might these expressive capabilities transform user interaction compared to traditional, non-expressive devices? (feel free to sketch if you want)

5. (10 points) **LA Feedback**

Your LA would appreciate having your feedback, as they use it to complete their final project by indicating where and how they improved and/or have more room for improvement.

Please take a few minutes to fill out this survey about how your LA did this quarter. Link: <https://tinyurl.com/LA-Feedback-S25>

(Please attach a screenshot as proof of completion)